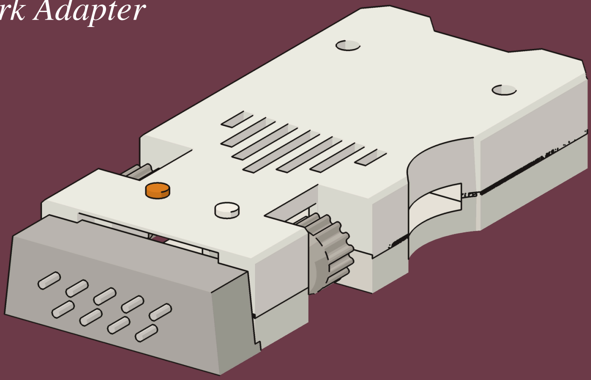
FujiNet

Personal Computer
Hardware Reference
Library

Guide to Operations

The RS-232 FujiNet Network Adapter



FN-RS232-GTO-001

A WORD ABOUT THIS GUIDE

This Guide is a labor of love by the worldwide FujiNet community. It is free, as is everything it describes: the FujiNet firmware, the CONFIG program, the MS-DOS drivers and utilities, and this very book. You are warmly encouraged to copy it, print it, and pass it to a friend.

The FujiNet community makes no promise that the network will be polite, that the weather report you fetch will be accurate, or that you will get any sleep once you discover how much there is to explore. The hardware is warranted only to the extent that the people who designed it want you to have a good time. If something does not work, the community will do its best to help — see **Getting Help** at the end of this Guide.

A NOTE ON RADIO INTERFERENCE

This equipment uses a small radio to reach your wireless network. It has been designed to be a good neighbor on the air. In the rare event that it disagrees with a nearby television or radio, moving the FujiNet, the computer, or the offended appliance a few inches usually restores the peace — the same cure the original Personal Computer manual recommended forty-odd years ago, and just as good today.

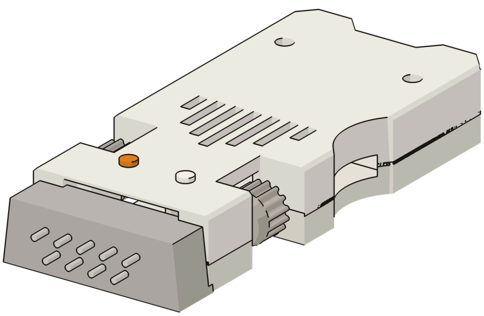
First Edition (2026)

FujiNet

Personal Computer Hardware Reference Library

Guide to Operations

The RS-232 FujiNet Network Adapter for MS-DOS



The FujiNet Community · fujinet.online

First Edition (2026)

The drawings of the Personal Computer and its rear panel in this Guide are stylized illustrations, drawn for clarity. The drawings of the FujiNet are rendered from the published FujiNet RS232-Rev1b hardware models. Your unit may differ in small details as the design improves.

Throughout this Guide, the names of the FujiNet drivers, utilities, and CONFIG screens are reproduced exactly as they appear on your computer, so that what you read here matches what you see there.

FujiNet is free and open. The firmware, the CONFIG program, the MS-DOS drivers and utilities, the hardware design, and this manual are all released under free-software and open-hardware licenses. Sources for everything live at `github.com/FujiNetWIFI` .

Preface

This Guide introduces you to the **RS-232 FujiNet** — a small adapter that plugs into the serial port of an IBM Personal Computer, or any close compatible, and gives it three things the original machine never dreamed of: virtual disk drives that hold their contents on the internet, a doorway onto your wireless network, and a printer that files its pages away as neat documents.

This publication is intended for anyone who will be setting up and using a FujiNet, whether or not they have ever used one before. No prior knowledge of networks is assumed. If you can switch on your computer and put a diskette in a drive, you already know enough to begin.

This Guide has six sections:

- **Section 1. Introduction** describes what the FujiNet is, what comes with it, and the handful of things you supply yourself.
- **Section 2. Setup** takes you from the box to a working FujiNet: connecting it, installing its two drivers, and running a short test to confirm all is well.
- **Section 3. Operations** is the heart of the Guide. It teaches the CONFIG program, the virtual disk drives, the network utilities, and the printer. New users should read it through; old hands may use it for reference.
- **Section 4. Problem Determination Procedures** helps you find and cure trouble, should any arise.
- **Section 5. Reference** collects the command summaries, the driver settings, and the list of network protocols in one place.
- **Section 6. Relocate** tells you how to move your system safely.

The Easy Way to Start

The fastest way through this Guide is to follow the black tabs printed along the edge of the right-hand pages. Each marks a major section, and together they trace the path from an unopened box to a computer that is on the network.

- **SETUP** — connect the FujiNet to your serial port, install its two drivers, and prove it is working.
- **OPERATIONS** — the learning-and-reference section. Read it once and you will know how to join your network, load disks, copy files, and print. Behind it sit smaller tabs for the parts you will reach for often: CONFIG, DISKS, NETWORK, and PRINTER.
- **PDPs** — the Problem Determination Procedures. If anything misbehaves, turn here first; the steps are written to be followed in order, top to bottom.
- **REFERENCE** — every command and setting, gathered for the day you need a quick reminder.

Note: If you are the sort of person who likes to switch a thing on and see what it does, skip to **Section 2** now. The FujiNet is hard to hurt and easy to reset, and you can always come back here.

Contents

SECTION 1. INTRODUCTION

Welcome to the world of the FujiNet. Your new adapter turns the serial port on the back of your Personal Computer into a doorway — onto your wireless network, onto a worldwide library of software, and onto a printer that never runs out of paper. Plug it in, and a machine designed before the network existed joins the network anyway.

In spite of its reach, the FujiNet is simple to operate, and *you* decide how technical you want it to be. At the simplest level you load its CONFIG program, pick a disk off a menu, and start computing. Everything else this Guide describes is there for the day you want it.

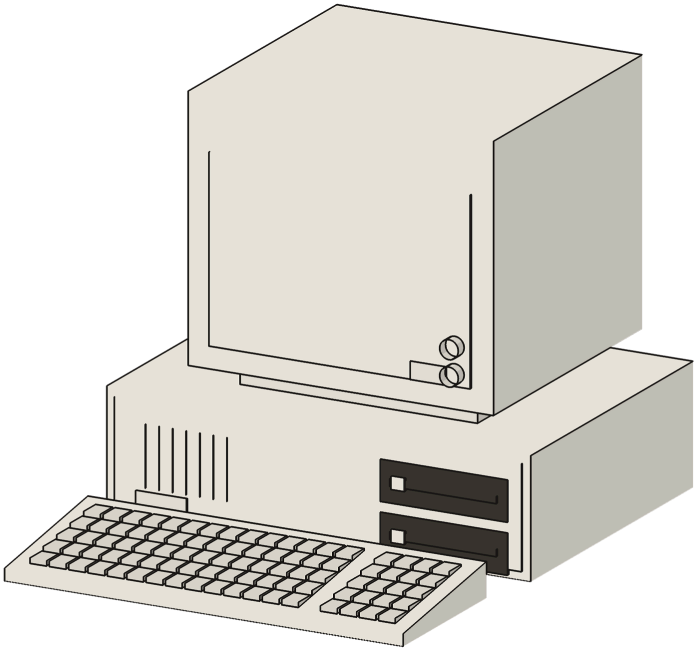
What the FujiNet Does

The FujiNet wears three hats at once:

- **Virtual disk drives.** The FujiNet pretends to be a stack of ordinary diskette drives. Where a real drive reads a floppy, the FujiNet reads a **disk image** — an exact copy of a diskette, kept as a file on a memory card or out on the network. DOS cannot tell the difference, and does not need to: as far as it knows, a drive answered the call. These appear as new drive letters, `C:`, `D:`, and beyond.
- **A network adapter.** A small radio inside the FujiNet joins your household wireless network. Programs written for the FujiNet can read the news, fetch the weather, call bulletin boards, and play games against real people across the world. A set of plain-DOS utilities — `NCOPY`, `NGET`, `NPUT`, and `FNSHARE` — let any computer copy files to and from the network without special software.
- **A printer.** The FujiNet can stand in for a printer on `LPT1:`. Anything your programs print is captured by the FujiNet and filed away as a finished document you can collect later.

Configuration Examples

A typical setup is shown below: an IBM Personal Computer with its display and keyboard, and the FujiNet on the serial port at the rear. No card to install, no case to open.



The Personal Computer — System Unit, Display, and Keyboard. The FujiNet attaches to the serial port on the back of the System Unit.

What You Supply

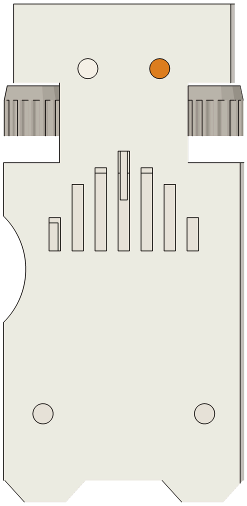
The FujiNet supplies the network. You supply the rest, and you very likely have it already:

- A **Personal Computer** — an IBM PC, PC/XT, PC/AT, or any close compatible — running **MS-DOS** (or PC DOS) version 3.0 or later.
- A free **serial port** (a COM port). Most computers have one as a 9-pin connector on the back; some older machines have a 25-pin connector, for which an inexpensive 9-to-25-pin adapter is used.
- A source of **USB power** for the FujiNet — a common USB wall charger, or a USB port on the computer. The FujiNet does *not* draw its power from the serial port.
- A **2.4-gigahertz wireless network** and its password.
- Optionally, a **microSD memory card** (64 GB or smaller, formatted FAT32) to hold a disk library of your own.

Note: The FujiNet's radio speaks the 2.4-gigahertz band only — the band every router still provides. If your network hides its 2.4-gigahertz band behind the same name as a 5-gigahertz band and the FujiNet has trouble joining, give the slower band its own name in the router's settings.

A Tour of the FujiNet

Take a moment to get acquainted before you plug anything in. Hold the FujiNet with its label up and its silver connector pointing away from you.



The FujiNet, seen from above. The serial connector is at the top; the reset button is at the lower edge.

- **The serial connector** — the silver 9-pin D-connector at the front. This is the plug that mates with your computer's serial port. A **knurled thumbscrew** stands at each side of it; finger-tight, they hold the FujiNet firmly to the computer.
- **The WiFi lamp (white)** — on the top face near the connector. It glows steadily once the FujiNet has joined your wireless network.
- **The bus lamp (orange)** — beside the WiFi lamp. It flickers when the computer and the FujiNet are talking — the FujiNet's version of the "in use" light on a diskette drive.
- **Button A** — a small button on the top face. You will use it only when updating the FujiNet's firmware; in everyday use it is left alone.
- **The Reset button** — at the rear edge, opposite the connector. Pressing it restarts the FujiNet itself. It does *not* restart your computer, and it is perfectly safe to press.
- **The USB-C jack** — on the rear edge. This is where the FujiNet takes its power, and how new firmware is loaded.
- **The microSD card slot** — on the rear edge. A memory card slides in here, contacts up; push to seat it, push again to release. The card is optional, but it gives you a disk library that needs no network at all.

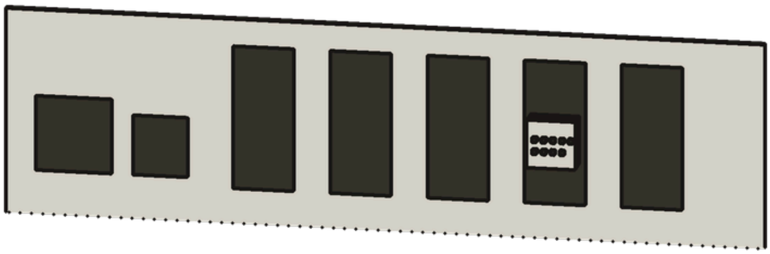
SECTION 2. SETUP

This section takes you from an unopened box to a working FujiNet. There are four short jobs: connect the adapter, copy its drivers onto a DOS diskette, tell DOS to load them, and switch on. Take them in order and you will be on the network in a few minutes.

CAUTION: Switch the computer OFF before connecting or disconnecting the FujiNet from the serial port. The serial port is not designed to be plugged and unplugged with the power on.

Connecting the FujiNet

1. **Find the serial port.** Look at the back of the System Unit for a **9-pin D-shaped connector** with the pins showing — this is the serial port, also called the COM port. On a computer with more than one, the first is **COM1**.



The back panel. The serial (COM) port is one of the D-shaped connectors among the option-card openings.

2. **Mate the connectors.** With the computer **off**, push the FujiNet's connector squarely onto the serial port. It fits only one way. Turn the two knurled thumb-screws finger-tight to hold it — snug is plenty; they are not meant to be forced.

Note: If your computer's serial port has **25 pins** instead of 9, fit an inexpensive 9-pin-to-25-pin serial adapter between the FujiNet and the port. Any electronics counter has them. Wire the FujiNet to a 9-pin end.

3. **Supply power.** Connect a USB-C cable from the FujiNet to a USB wall charger or a USB port. The lamps on top flicker as it starts up, then the white WiFi lamp waits, unlit, until the FujiNet has a network to join.

4. That is the whole hardware installation. There is no card to seat and no case to open. Leave the FujiNet powered; the rest of the work is done in software.

The FujiNet Tools Diskette

The FujiNet's drivers and utilities arrive as a small collection of files — the **FujiNet Tools**. You will find them on the diskette image supplied with your FujiNet, or as a download from `fujinet.online` . The collection holds:

File	What it is
FUJINET.SYS	The disk-drive driver — loads in CONFIG.SYS
FUJIPRN.SYS	The printer driver — loads in CONFIG.SYS
CONFIG.EXE	The CONFIG program — the menu you live in
FMOUNT.EXE	Mounts and ejects disk images from DOS
NCOPY.EXE	Copies files to and from the network
NGET.EXE	Downloads one file from a network address
NPUT.EXE	Uploads one file to a network address
FNSHARE.EXE	Maps a network share to a drive letter

1. Copy these files onto your everyday **DOS system diskette** (or your hard disk), the same disk you start the computer from. They take very little room. If the FujiNet came with a ready-made bootable diskette image, you may simply start the computer from it instead.

Note: The FujiNet provides its drives over the serial cable, but DOS itself must come from somewhere the computer can already start from — a diskette or a hard disk. The FujiNet's drives appear *after* DOS is running and the drivers are loaded.

Installing the Drivers

Two files on your start-up disk tell DOS what to do at switch-on: `CONFIG.SYS` and `AUTOEXEC.BAT`. Add the FujiNet to each with any text editor.

1. Add these two lines to `CONFIG.SYS`. The first loads the disk driver; the second loads the printer driver:

```
DEVICE=FUJINET.SYS FUJI_PORT=1 FUJI_BPS=115200
DEVICE=FUJIPRN.SYS
```

2. Add this line to the end of `AUTOEXEC.BAT` so the CONFIG program greets you at every start-up:

```
CONFIG.EXE
```

The two settings on the `FUJINET.SYS` line are worth knowing:

- `FUJI_PORT` names the serial port: `1` for COM1, `2` for COM2, and so on. Set it to the port you used. (For an unusual port you may give an address and interrupt directly, as in `FUJI_PORT=0x2F8,3` .)
- `FUJI_BPS` sets the speed, in bits per second. `115200` is the standard, and matches a FujiNet as it ships. Both ends must agree; if you ever change the FujiNet's speed, change it here to match.

Note: Leave the two driver lines near the top of `CONFIG.SYS`. If you prefer the computer not to set its clock from the FujiNet, add the word `NOTIME` to the `FUJINET.SYS` line.

First Power-On

With the FujiNet connected and powered, and the drivers installed, restart the computer. Watch the screen as DOS starts:

1. As `CONFIG.SYS` is read, the FujiNet driver announces itself. The exact numbers will differ, but the shape of the message is always the same:

```
FujiNet driver 0.8 Open Watcom 2.0 on MS-DOS 6.2
```

2. Seeing that line means `FUJINET.SYS` loaded, found the FujiNet across the serial cable, and added a set of new disk-drive letters — beginning at `C:` (or the next free letter after your real drives). If you asked it to, the FujiNet also quietly set the computer's clock to the correct date and time, fetched from the internet.
3. When `AUTOEXEC.BAT` reaches `CONFIG.EXE`, the CONFIG program fills the screen. The very first time, it begins by scanning for wireless networks (Section 3). After that, it opens straight to its main menu.

Note: If the driver prints a message that it could *not* find the FujiNet, the computer started up normally but no FujiNet drives were added. Turn to **Section 4, Problem Determination Procedures** the usual causes are the wrong COM port or a loose connector.

The Mini-Test

Here is a thirty-second check that everything is talking. Leave CONFIG (or, if it has not started, work at the DOS prompt) and type:

```
FMOUNT
```

The FujiNet answers with a list of the drive letters it has provided, and what is in each — the disk equivalent of a roll-call:

```
--- C: R 2:DISK1.IMG
    D: -- no disk selected --
    E: -- no disk selected --
```

If you see your FujiNet drive letters listed, the adapter, the cable, the serial port, and the drivers are all working together. Your FujiNet has passed its self-test, and you are ready for Section 3.

Summary

This completes Setup. You have connected the FujiNet to the serial port, given it power, copied its drivers onto your start-up disk, told DOS to load them, and confirmed that the computer and the FujiNet are talking.

From now on, every time you switch on, the FujiNet's drives appear by themselves and the CONFIG program greets you. You are ready to join your network and load your first disk.

SECTION 3. OPERATIONS

This section is the heart of the Guide. It teaches the CONFIG program — the menu you will use to join your network and load disks — and then the plain-DOS utilities that copy files, map network shares, and print. New users should read it through once; afterward it serves as a handy reference.

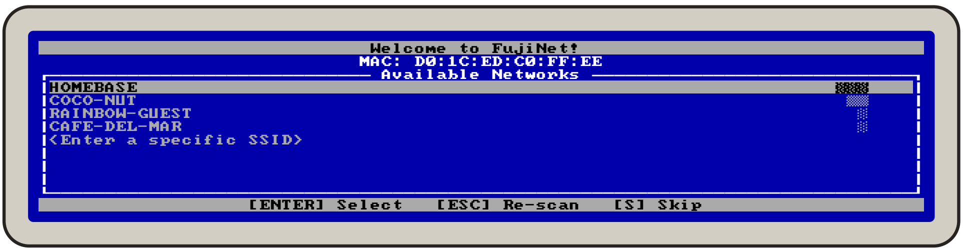
Getting CONFIG Ready

CONFIG runs whenever `CONFIG.EXE` is started — automatically at every switch-on, or by typing `CONFIG` at the DOS prompt. It fills the screen with a tidy display in the familiar style of the DOS Editor: a title bar across the top, your working area in the middle, and a **status bar** across the very bottom that always lists the keys you can press right now.

Note: Throughout CONFIG, the keys you can press are shown in the bottom status bar inside square brackets, like `[ENTER]` and `[ESC]`. When in doubt, read that bottom line: it always tells you what to do next.

Joining Your Network

The first time CONFIG runs, it scans the airwaves and lists every wireless network it can hear. The signal-strength bars at the right show which are near (▓▓▓▓) and which are far (░).

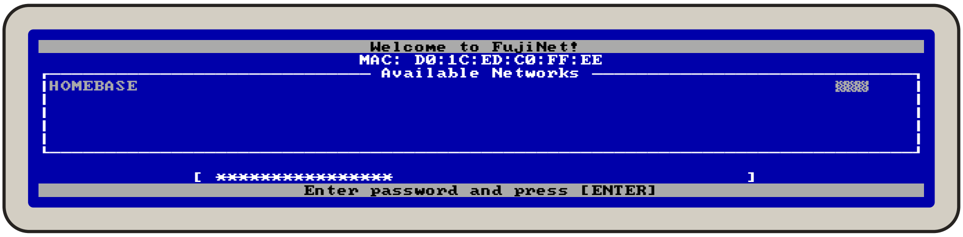


The network scan. The bright bar marks your place; the arrow keys move it.

To join a network:

- Press ↑ and ↓ to move the bright bar to your network, then `[ENTER]`.
- If your network does not broadcast its name, move to `<Enter a specific SSID>`, press `[ENTER]`, and type the name.
- Press `[ESC]` to scan again, or `[S]` to skip the network for now and set it up later.

When you choose a network, CONFIG asks for the password. Type it carefully — capitals count — and press `[ENTER]`. Each character shows as an asterisk:

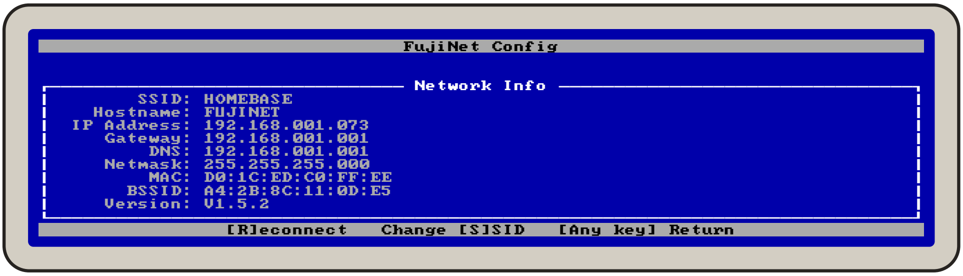


Entering the network password.

Press **[ENTER]** and the FujiNet joins up. The white WiFi lamp comes on and stays on. Your network name and password are remembered inside the FujiNet, so from now on it reconnects by itself, before the screen has even warmed up.

The Configuration Screen

To see the FujiNet's vital signs at any time, press **[C]** from the main screen. CONFIG shows the network it has joined, the address it was given, and its firmware version:

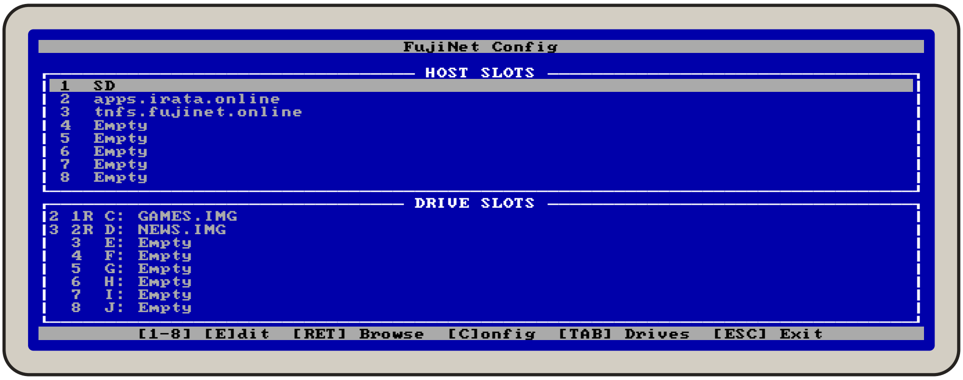


The Configuration screen. Note the IP Address line.

Note: Note the **IP Address** line. While the computer is on, the FujiNet serves a full settings page to any web browser in the house. From a modern computer or telephone, visit that address (for example `http://192.168.1.73`) to rename the device, manage WiFi, choose printer styles, and update firmware from a comfortable chair.

Host Slots and Drive Slots

Once you are on the network, CONFIG shows its main screen. It is two short lists, one above the other, and once you can read them you can do everything.



The main screen: Host Slots above, Drive Slots below.

A **host** is any place disk images live: a library on the internet, a file server on your own network, or a microSD card (always called `SD`). The FujiNet remembers eight of them — the **host slots**.

A **drive slot** is one of the disk drives your computer sees. Loading a disk image into a drive slot is the FujiNet's way of sliding a diskette into a drive — and each drive slot shows the DOS drive letter it answers to (`C:`, `D:`, and so on).

On the Host Slots

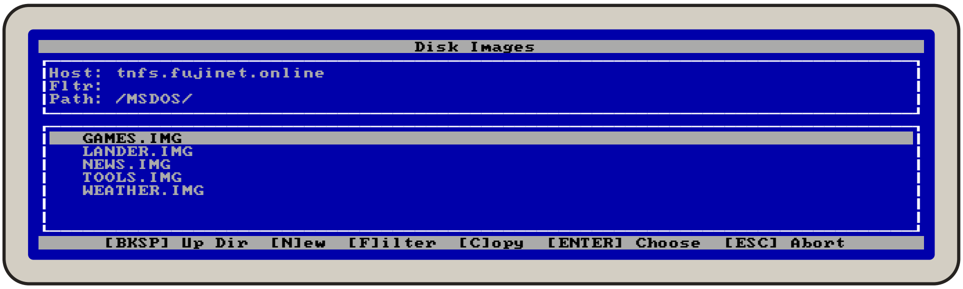
- ↑ ↓ move the bright bar; `[1]`–`[8]` jump straight to a slot.
- `[E]` edits the highlighted slot — type a host name (up to 32 characters) and press `[ENTER]`.
- `[ENTER]` opens the highlighted host so you can browse it.
- `[TAB]` switches down to the Drive Slots list.
- `[ESC]` leaves CONFIG and starts computing.

Out of the box, slot 1 is `SD` and another slot points at a public library. Worthwhile libraries to type into an empty slot include:

- `tnfs.fujinet.online` — the community's main library
- `apps.irata.online` — applications and on-line services

Loading a Disk

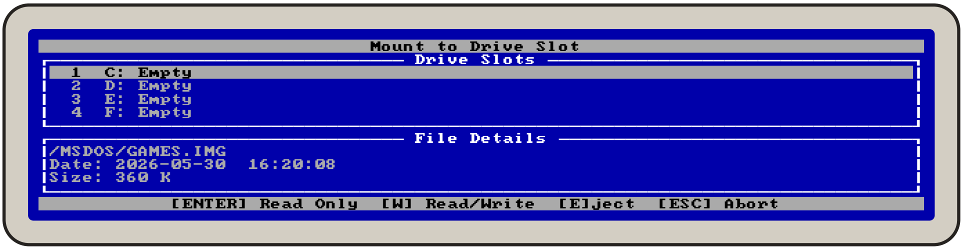
Move the bar to a host, press `[ENTER]`, and CONFIG opens its catalog. Names ending in `/` are folders; step into the one for your machine.



Browsing a library. Folders end in a slash; disk images do not.

- ↑ ↓ move the bar. Keep going past the bottom and the next page of names slides in.
- `[ENTER]` opens a folder, or chooses a disk image.
- `[BKSP]` backs out to the folder above.
- `[F]` types a filter, like `W*.IMG`, to show only matching names. An empty filter clears it.

Put the bar on a disk image, press `[ENTER]`, and CONFIG asks which drive slot it should go in, showing the file's details while you decide:



Choosing a drive slot for the disk.

- ↑ ↓ choose a drive slot.
- `[ENTER]` loads it **read-only** — nothing can change it, like a diskette with the write-notch covered.
- `[W]` loads it **read/write** — programs may save onto the image.
- `[E]` ejects whatever is in the slot.

Note: Public libraries do not allow writing, so load their disks read-only — the `[ENTER]` key. Save `[W]` for disks on your own SD card or your own server.

Leaving CONFIG

When your disks are loaded, press `[ESC]`. CONFIG announces `Mounting all slots...` and hands control back to DOS. Your loaded disks are waiting at their drive letters — type `DIR C:` and see.

New Disks and Copies

While browsing a host you can write to (your SD card, say), press `[N]` to make a brand-new, blank disk image. CONFIG offers the familiar PC diskette sizes — `[1] 360K`, `[2]` `720K`, `[3] 1.2MB`, `[4] 1.44MB` — then asks for a name. The new image appears in the folder, blank and ready to format with the DOS `FORMAT` command.

To keep a copy of something from a network library, highlight it and press `[C]`. CONFIG asks which host to copy *to* — choose `SD` — lets you pick the folder, and copies the file across with no help from the computer's memory at all.

The Drive Letters

Each FujiNet drive slot answers to an ordinary DOS drive letter. The driver hands out the next free letters after your real drives — usually `C:`, `D:`, and onward — and they behave exactly like diskette drives. Once a disk image is loaded into a slot, you can:

```
DIR C:
COPY C:*.* D:
A:\> C:
C:\> _
```

There is nothing new to learn. **`DIR`**, **`COPY`**, **`TYPE`**, **`FORMAT`**, and the rest work on FujiNet drives just as they do on real ones. A disk loaded *read/write* can be written to; a *read-only* disk politely refuses, the same as a diskette with its notch covered.

Mounting Disks from DOS

You need not return to CONFIG to manage disks. The **`FMOUNT`** command does the same jobs from the DOS prompt. Typed alone, it lists your FujiNet drives and what is in each:

```
C:\> FMOUNT
▬▬▬ C: R 2:GAMES.IMG
-▮- D: R 3:NEWS.IMG
    E: -- no disk selected --
```

The little mark at the left tells the state of each drive: three bars for a disk that is loaded and ready, a barred dash for one chosen but not yet mounted, and blank for an empty slot. With options, **`FMOUNT`** loads and ejects:

Command	What it does
`FMOUNT -a`	Mount every slot that has a disk chosen
`FMOUNT C:`	Mount the disk chosen for drive C:
`FMOUNT -w C:`	Mount drive C: read/write
`FMOUNT -e C:`	Eject the disk in drive C:
`FMOUNT -t 2`	Show which drive letter slot 2 became

The Network Utilities

Three small programs let any DOS program — or you, by hand — reach across the network without knowing a thing about it. They address the network through **N: names**: a protocol, a colon, and an address, such as **`N:HTTP://`** or **`N:TNFS://`** . The protocols the FujiNet understands are listed in Section 5.

NGET — fetch one file

Give it a network address and a name to save under:

```
C:\> NGET N:HTTP://example.com/files/manual.txt MANUAL.TXT
      4096 bytes transferred.
```

NPUT — send one file

The mirror image — a local file, then where to put it:

```
C:\> NPUT REPORT.TXT N:TNFS://192.168.1.10/uploads/REPORT.TXT
```

NCOPY — a file-copying conversation

For more than one file, `NCOPY` opens an interactive session on a host. At its `ncopy>` prompt you can list, change folders, and copy in both directions:

```
C:\> NCOPY N:TNFS://tnfs.fujinet.online/
ncopy> dir
MSDOS/         <DIR>     2026-May-01 09:14
README.TXT       812     2026-Apr-22 18:03
ncopy> cd MSDOS
ncopy> get TOOLS.IMG TOOLS.IMG
ncopy> quit
```

The commands are few and plain: `dir` (or `ls`) to list, `cd` to change folder, `get` to fetch a file, `put` to send one, and `quit` to finish. If the host needs a name and password, `NCOPY` asks for them.

Mapping a Network Share

`FNSHARE` goes one step further: it makes a whole network folder appear as a DOS drive letter, so `DIR`, `COPY`, and your programs can use it as if it were a local disk. It stays resident after you run it.

```
C:\> FNSHARE map L: N:TNFS://192.168.1.10/shared
FujiNet installed as L:
```

From then on, `L:` is the shared folder. If the server asks for a name and password, `FNSHARE` prompts for them before mapping the drive.

Note: `FNSHARE` maps a **folder of ordinary files** to a drive letter, which is perfect for documents and programs. To run software from a **disk image**, load it into a drive slot with CONFIG or `FMOUNT` instead.

The Printer

The second driver, `FUJIPRN.SYS`, gives you a printer that never jams, never runs dry, and files every page away for you. Once it is loaded, anything the computer sends to `LPT1:` — the first printer — goes to the FujiNet instead of out a cable.

Use it exactly as you would a real printer:

```
C:\> COPY README.TXT LPT1:
C:\> PRINT REPORT.TXT
```

You may also press `Shift-PrtSc` to print the screen, or use the `Print` command in your favorite program. Whatever you send, the FujiNet collects and turns into a tidy document — by default, a PDF you can read or print on a modern machine.

1. Print from any program, the ordinary way.
2. Open the FujiNet's web page (the address on the Configuration screen) in a browser on a modern computer or telephone.
3. Find your printout waiting there, ready to view, save, or print on a real printer.

Note: The FujiNet can imitate several kinds of printer — a plain-text page, a dot-matrix style, and others — and you choose which from its web page. For everyday DOS text, the standard setting needs no attention at all.

Using DOS with the FujiNet

There is little to add: a FujiNet drive *is* a DOS drive. You can make one of them the current drive, run programs from it, save your work onto a read/write disk, and switch between disks by loading different images into the slots. A few friendly reminders:

- Load a disk **read/write** before expecting to save onto it. Read-only is the safe default, and the usual choice for software you have fetched from a public library.
- A blank image you made with `[N]ew` is truly blank — give it a file system with `FORMAT` before first use, the same as a fresh diskette.
- When you are finished, there is no need to "eject" anything. Switch off when you like; your disk images are safe on their card or server.

SECTION 4. PROBLEM DETERMINATION PROCEDURES

When something does not behave, this section helps you find the cause and cure it. Work through the steps for your symptom in order, top to bottom, and stop as soon as the trouble clears.

Start Here

Before consulting the tables, check the three things that cause most trouble:

1. **Power.** Is the FujiNet's USB cable connected to a live source? The lamps on top should light when it starts.
2. **The connector.** Is the FujiNet pushed fully onto the serial port, with both thumbscrews snug? A connector half-seated is the commonest fault of all.
3. **The port.** Does the `FUJI_PORT` setting in `CONFIG.SYS` name the port you actually used — `1` for COM1, `2` for COM2?

If all three are sound and trouble remains, find your symptom below.

The Symptom Tables

Symptom: At start-up the driver says it cannot find the FujiNet.
The driver loaded but got no answer over the cable. Switch off, and check that the FujiNet is powered and firmly connected. Confirm `FUJI_PORT` names the right COM port. If you set an unusual speed, make sure `FUJI_BPS` matches the speed the FujiNet expects (the standard is `115200`). Then switch on again.

Symptom: No new drive letters appear, and no message from the driver.
The driver line is not being read. Check that `DEVICE=FUJINET.SYS` is in `CONFIG.SYS`, spelled correctly, and that `FUJINET.SYS` is really on the start-up disk. Restart and watch for the driver's banner.

Symptom: Characters on the FujiNet drives are garbled, or files copy wrong.
The two ends are talking at different speeds, or the serial port is unreliable. Make `FUJI_BPS` match the FujiNet's speed. If trouble persists, try a slower speed at both ends (for example `19200`), or a different serial port.

Symptom: CONFIG will not join the wireless network.
Check the password — capitals count. Make sure the network offers a **2.4-gigahertz** band; the FujiNet cannot use 5-gigahertz-only networks. Move the FujiNet nearer the router, or press `[ESC]` to re-scan and pick a network with stronger signal bars.

Symptom: A host slot will not open, or shows no files.
Check the host name for a typo. A public server may be busy or down; the current list of working servers is kept at the FujiNet web site. Confirm the white WiFi lamp is lit — without a network, no host can be reached.

Symptom: A program cannot save to a FujiNet drive.
The disk is loaded read-only. In CONFIG, put the bar on the drive in the Drive Slots list and press `[W]` to switch it to read/write; or from DOS, `FMOUNT -w` the drive. Public-library disks cannot be made writable — copy one to your own card or server first.

Symptom: Printer output never appears on the FujiNet web page.
Confirm `DEVICE=FUJIPRN.SYS` is in `CONFIG.SYS` and loaded. Make sure your program prints to `LPT1:`. Give the FujiNet a moment to finish the document, then refresh its web page.

Symptom: A FujiNet drive letter clashes with another disk or network drive.
Another driver claimed the same letters. Load `FUJINET.SYS` earlier in `CONFIG.SYS`, or adjust the other driver, so each gets its own range of letters. Use `FMOUNT` to see which letters the FujiNet received.

Note: If a symptom is not listed here, or a cure does not work, the community is glad to help. See **Getting Help** at the back of this Guide — bring the exact wording of any message you saw.

SECTION 5. REFERENCE

This section gathers the settings and commands in one place, for the day you need a quick reminder.

Driver Settings

These go on the `DEVICE=FUJINET.SYS` line in `CONFIG.SYS`:

Setting	Default	Meaning
`FUJI_PORT`	`1`	Serial port: 1–4, or an address and IRQ such as `0x2F8,3`
`FUJI_BPS`	`115200`	Speed in bits per second; must match the FujiNet
`NOTIME`	—	If present, do not set the DOS clock from the FujiNet

The CONFIG Keys

Key	Action
↑ ↓	Move the bright bar through a list
`1–8`	Jump straight to a slot
`ENTER`	Choose, open, or (on a disk) load read-only
`W`	Load a disk read/write
`E`	Edit a host slot, or eject a drive slot
`TAB`	Switch between the Host and Drive lists
`C`	Show the Configuration (Network Info) screen
`N`	Make a new, blank disk image
`F`	Filter the file list
`BKSP`	Go up one folder
`ESC`	Leave CONFIG, mounting all slots

Command Summary

Command	Purpose
`CONFIG`	Start the full-screen CONFIG program
`FMOUNT`	List FujiNet drives and what is loaded
`FMOUNT -a`	Mount every slot that has a disk chosen
`FMOUNT -w C:`	Mount drive C: read/write
`FMOUNT -e C:`	Eject the disk in drive C:
`FMOUNT -t 2`	Show the drive letter for slot 2
`NGET src dest`	Download one file from an N: address
`NPUT src dest`	Upload one file to an N: address
`NCOPY host`	Open an interactive copy session on a host
`FNSHARE map L: url`	Map a network share to drive L:

Inside `NCOPY`: `dir` or `ls` to list, `cd` to change folder, `get` and `put` to copy, `quit` to finish.

Network Protocols

The network utilities reach the world through **N: names** — a protocol, a colon, and an address. The FujiNet understands these protocols; any of them may be used wherever an N: address is asked for:

Name	Reaches
`N:HTTP://`	Web servers (also `HTTPS://` for secure sites)
`N:TNFS://`	TNFS servers — the 8-bit world's disk libraries
`N:FTP://`	File-transfer servers
`N:SMB://`	Windows and Samba shared folders
`N:NFS://`	Unix network-file-system shares
`N:TCP://`	A raw connection to any TCP service
`N:UDP://`	Raw datagrams to any UDP service
`N:SSH://`	A secure shell to another computer
`N:TELNET://`	Terminal sessions on bulletin boards and hosts

Note: `FNSHARE` and `NCOPY` work with the file-serving protocols — `TNFS`, `FTP`, `SMB`, `NFS`, and `HTTP`. The raw protocols (`TCP`, `UDP`, `TELNET`, `SSH`) are there for `NGET`, `NPUT`, and for programs written to use the network directly.

Disk Image Sizes

When you make a new disk image with `[N]ew`, CONFIG offers the standard PC diskette formats. After creating one, give it a file system with the DOS `FORMAT` command, the same as a new diskette.

Choice	Format
`[1]`	360 KB — 5.25-inch double-density
`[2]`	720 KB — 3.5-inch double-density
`[3]`	1.2 MB — 5.25-inch high-density
`[4]`	1.44 MB — 3.5-inch high-density
`[C]`	Custom — your own sector count and size

SECTION 6. RELOCATE

Whenever you need to move your system, a few moments' care saves trouble later. The FujiNet asks little: it is small, light, and has nothing inside that minds being carried.

Moving the FujiNet

1. Switch the computer **off**.
2. Loosen the two thumbscrews and lift the FujiNet straight off the serial port. Unplug its USB power.
3. If a microSD card is fitted, you may leave it in place; it is held securely. Your disk images on it travel with it.
4. Pack the FujiNet where it will not be crushed. There are no moving parts and no diskettes to fall out.

Taking It With You

The FujiNet remembers the last network it joined. In a new place, with a different wireless network, simply press `[C]` in CONFIG, then `[S]` to change the network, and choose the new one from the scan — exactly as you did the first time (Section 3). Everything else carries on as before.

Note: Away from any network, the FujiNet still serves disk images from its microSD card. A library on the card needs no internet, no router, and no password — handy for a club meeting, a class, or a long trip.

SECTION 7. GETTING HELP

The FujiNet is the work of a worldwide community of enthusiasts, and that community is friendly and quick to help. If this Guide has not answered your question, here is where to turn:

- **The FujiNet web site** — `fujinet.online` . Downloads, documentation, the firmware updater, and a current list of public libraries.
- **The FujiNet chat server** — a link is on the web site. Ask a question and a real person usually answers within the hour.
- **The FujiNet users' group** — for show-and-tell, tips, and troubleshooting among fellow owners.
- **The source code** — `github.com/FujiNetWIFI` . Everything that makes the FujiNet work is open for you to read, learn from, and improve.

When you ask for help, it speeds things along to mention: which computer and version of DOS you use, which serial port and speed, the firmware version from the Configuration screen, and the exact wording of any message you saw.

Welcome to the network. We are glad you are here.

FujiNet

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